Gas Absorption in a Packed Bed
Pre-Lab Plan 1
Chem Engr Lab II
Date: 23 Jan 2020

Commented [PTH1]: 31.5/35

Team 3, Thursday, PM section
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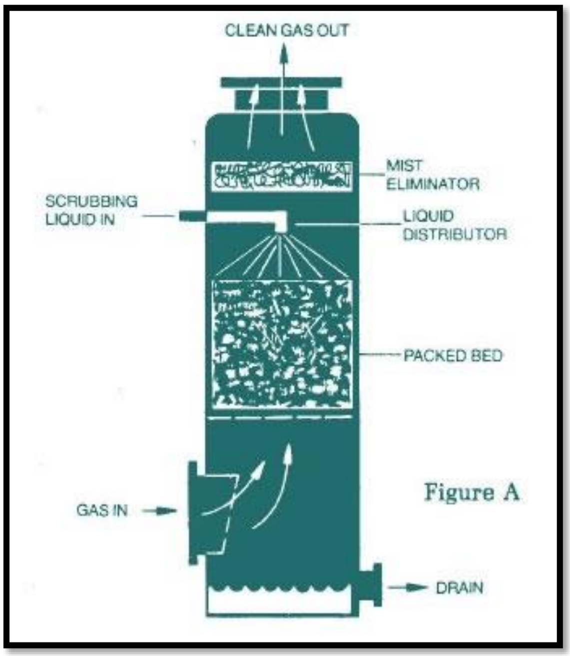

Figure A

Commented [AZ2]: Nice picture

1. **Experimental Objective**

Commented [PTH3]: 4/5

Commented [PTH4]: Missing info here. There are three parts to this experiment and you should have summarize all of them in the objective in another sentence or two.

The objective of this experiment is to determine the characteristics of fluid flow, flooding point, and the overall mass transfer coefficient in the removal of CO_2 from an air stream with water, using a packed bed gas absorber.

2. Experimental

The equipment use for this experiment is an Armfield UOP7 Gas Absorption Column filled with 1-cm Raschig rings (1). The apparatus consists of a cylindrical column with a gas inlet and a distribution space at the bottom, and a liquid inlet and distribution space at the top, with gas and liquid outlets respectively. All streams have rotameters (9, 10, 11) to determine the volumetric flowrate. The absorber system is constructed such that the gas flows countercurrent to the liquid. Furthermore, the system has two water manometers (2) to determine the pressure drop across the absorber unit. The composition of the gas stream is determined volumetrically by using a Hempl Apparatus (12).

Commented [AZ5]: Good introduction to this section. Don't forget to introduce the figures and tables in the text before they appear as well.

2.1 Apparatus

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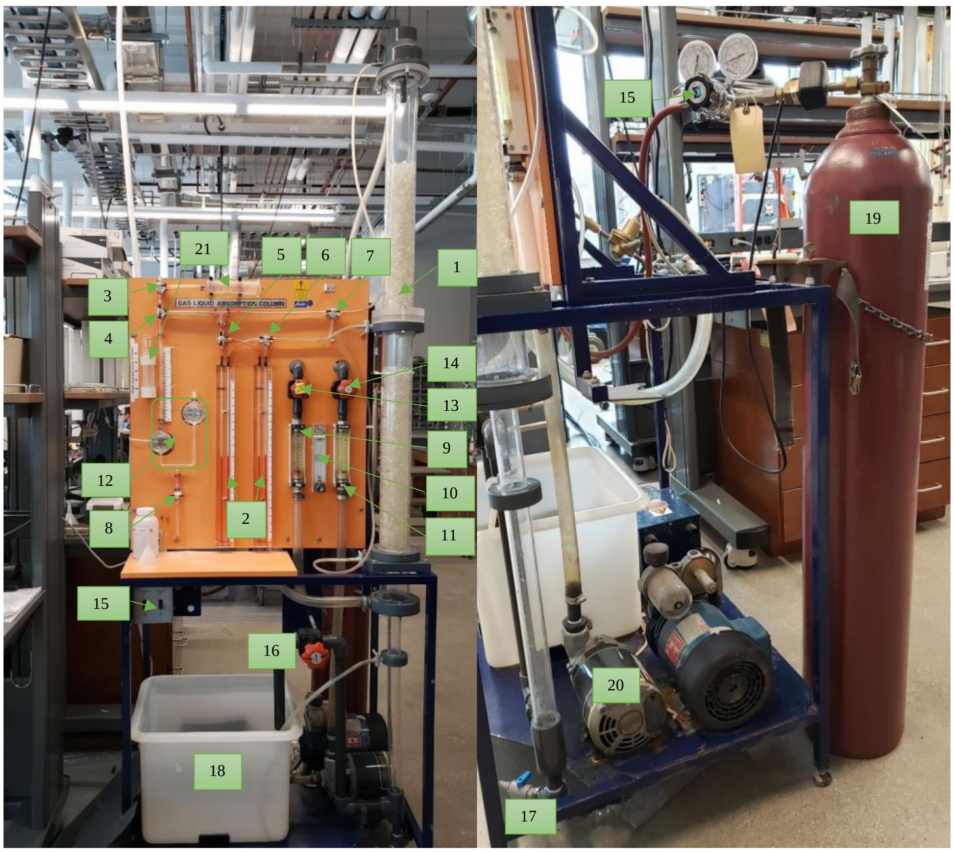

Figure 1. Armfield UOP7 Gas Absorption Column

Table 1. Apparatus Component and Description

No.	Component	Description
1	Gas absorption column	Filled with rings for gas absorption process
2	Manometer	To measure the column's pressure drop
3	Valve :1	To deliver sample to valve 2
4	Valve :2	To deliver sample to Hampel apparatus globes
5	Valve :3	To connect to manometer
6	Valve :4	To connect to manometer
7	Valve :5	To connect to the packed column to Hempel Apparatus
8	Valve :6	To drain the Hampel Apparatus
9	Rotameter	To alter the flow rate for air, enter in absorber
10	Rotameter	To alter the flow rate for CO2, enter in absorber
11	Rotameter	To alter the water flow rate, enter in absorber
12	Hempel Apparatus	To analysis the gas sample
13	Control valve (C1)	To allow air flow into the system
14	Control valve (C2)	To allow water flow into the top of the absorber
15	Control valve (C3)	To control CO2, enter the system
16	Control valve (C4)	To control water flow back to the water tanks
17	Drain Valve (D1)	Drain the water from packed bed absorber
18	Water tank	To fill the water to ¾ or full to flow throughout the system
19	CO2 gas tanks	High pressure tank support CO2 into the absorber
20	Pump	To pump the water through the system
21	Syringe	To suck out air out of the line to measure the volume

Commented [AZ7]: There is a valve to the right of 16 that is used to take water samples for titration analysis

Commented [AZ8]: Two manometers

Commented [AZ9]: Technically yes, but don't operate this valve, leave it always open. There is a needle valve in the back that is used to control air flowrate. Also a shutoff valve.

Commented [AZ10]: Two stage regulator and heater located here.

Commented [AZ11]: Just one cylinder

Commented [AZ12]: Better name is piston/cylinder

2.2 Material and Supplies

Commented [PTH13]: 4/5

Table 2. Materials and Supplies

No.	Material & Supplies	Description
1	Compressed Air	House air stream
2	Phenolphthalein indicator solution	To Add 3 to 5 drops to check if amount of free CO2 in the samples
3	Sodium Hydroxide Solution	0.05 N or 0.01 N and 1.0 N
4	Carbon Dioxide Gas	From air stream into water in the packed bed gas absorber
5	Deionized Water	To fill in the reservoir tank.
6	125 mL Erlenmeyer flasks	To collect samples
7	Stopwatch	To measure time
8	50 mL burette and burette clamp	To use for the titration
9	50 mL graduated cylinder	To collect samples and for measuring void space
10	Pasture pipet	To add the indicator
11	Thermometer	To measure the temperature of the water
12	Tap water	To measure void volume of packed bed
13	Basin	To collect water draining out of reservoir.
14	Squirt bottle of DI water	To wet the piston syringe

Commented [PTH14]: you're missing parafilm. You need it to cover your sample and prevent CO2 from escaping

Commented [PTH15]: you'll need stir plate and possibly stir bar

2.3 Experimental Plan

Commented [PTH16]: 10/10

The experiment consists of four objectives. First, the packed bed void volume will be measured. Then the packing and packed bed density will be calculated. The second part entails calculating the pressure drop across the packed bed. Thirdly, water and carbon dioxide will be brought into equilibrium and the carbon dioxide in the air leaving the column will be measured. Lastly, the absorption and desorption of carbon dioxide into the air and into water will be analyzed.

Procedure

Part 1: Characterization of Liquid and Gas Flow in Packed Columns

Packed Bed Flooding:

1. Determine the void volume of packed bed
    a. Use the equation $\in = \frac{V_{void}}{V_{apparent}}$ or $\in = 1 - \frac{\rho_B}{\rho_s}$ where ∈ is void volume defined as volume of voids/total volume of packed bed and ρ_s is the density of the solid and ρ_B is the bulk density of the packed particles.
    b. Measure the amount of water displaced by the packed bed particles using graduated cylinders.
    c. Calculate the density of the packing by dividing the mass by the solids of their displacement volume.
    d. Directly determine the density of the packed bed by measuring the volume occupied by a known weight of randomly packed solids.
2. Measure the pressure drop across the packed bed as a function of the air and water flow rates with the manometer and appropriate pressure taps.
    a. Close the drain valve under the liquid reservoir tank, the control valve C4, and the column drain valve D1.
    b. Fill the liquid reservoir tank at the base of the column three-quarters full with DI water.
    c. Record temperature of the water.
    d. Adjust the air flow rate through flowmeter F2 to 20 L/min using control valve C2, and allow air to flow through the column for 5 minutes to dry the packing.
    e. Measure the pressure drop across the bed for different gas flow rates with no liquid flow. Use flows 20, 40, 60, 80, 100, 120 L/min.
    f. Open control valve C4. With gas control valves C2 and C3 closed, start the liquid pump, open valve C1 a couple of turns for 15 to 30 seconds to expel air from the pump and adjust the water flow through the column to 1.5 L/min on flowmeter F1 by adjusting flow control valve C1.
    g. Open gas flow control valve C2 and adjust flowmeter F2 to 20 L/min. Measure the gas pressure drop as you increase the air flowrate until flooding occurs.
    h. Repeat for water flowrates of 3 L/min, 6 L/min, (and 8 L/min if needed) and record pressure drop.

Part 2a: Bringing Water to Equilibrium with CO2 and Analysis of CO2 in Water

1. Close drain valve under the liquid reservoir tank, the control valve C4 and the column drain valve D1.
2. Fill the liquid reservoir tank at the base of the column three-quarters full with DI water.

3. Record the temperature of the water.
4. Place cover on the tank.
5. Find the initial CO_2 concentration.
    a. Collect 50 mL with a graduated cylinder from the reservoir tank, purging the sample line before collecting the sample.
    b. Place sample in 125 mL Erlenmeyer flask.
    c. Add 3-5 drops of phenolphthalein indicator solution. If the sample is immediately red, no free CO_2 is present. If the sample is colorless, swirl the flask and titrate with standard NaOH solution. End point is reached when the solution turns pink for about 30 seconds.
    d. Record volume of NaOH added.
    e. Calculate the amount of free CO_2 from the equation.
6. Open valve C4.
7. With gas flow control valve C2 and C3 closed, start the liquid pump, open valve C1 a couple turns for 15-30 seconds to expel air from the pump.
8. Adjust water flow to the column to 6 L/min on flowmeter F1 by control valve C1.
9. Adjust the air flow rate through flowmeter F2 to 20 L/min using control valve C2, adjusting the air flow rate as needed to keep it at 20 L/min. Note any readjustments needed.
10. Open the main valve on the CO_2 cylinder all the way and slowly adjust pressure regulating valve on the CO_2 regulator to give a gauge reading of 20 psi. Adjust valve C3 to give a CO_2 flowrate on flowmeter F3 corresponding to the flowrate assigned (2 through 5 L/min).
11. After 10 min of operation, take 50 mL samples through S4 at 3-minute intervals until the dissolved CO_2 values are constant.
    a. Use the method described in Part 2a, number 5.
12. Measure the volume fraction of CO_2 in the air leaving the column (from S2) once the dissolved CO_2 levels are constant.
    a. First find the volume fraction of CO_2 in ambient air.
    b. Connect the compressed air line to the unit and ensure open the valve on the air line. Adjust the air flow rate through the flow meter to 20L/min using control valve C2 adjusting as needed to keep at 20 L/min.
    c. Reconnect the CO_2 line to the flow meter at the rear of the panel and adjust the flow to 2 L/min.
    d. Collect a sample of gas from S2
    e. Fill two globes of the absorption analysis equipment with 1.0 N NaOH solution using a small funnel and tubing.
    f. Adjust level on globes to "0" mark on the sight tube by draining liquid through valve Cv into a flask.
    g. After 15 minutes of steady operation, take samples of gas from S2.
    h. Pull piston of the syringe out to the stop. Take a squirt bottle of DI water and wet the piston of the syringe. Repeat this step as necessary since the water does evaporate.
    i. Flush the sample lines by repeated sucking from the line using the syringe and expelling the contents of the syringe to the atmosphere. Estimate the volume of

the tubing leading to the syringe by measuring inner diameter and length. Determine how many times needed to purge the line, around 5 – 6 times.

   j. Fill the syringe from the selected line by drawing the syringe piston out slowly. Note the volume taken into the syringe V1. It should be about 40 mL. Wait 2 minutes to allow the gas to come to the temperature of the syringe.
   k. Isolate the syringe from the column and the absorption globe and vent the syringe to the atmospheric pressure. Close after about 10 seconds.
   l. Connect the syringe to the absorption globe. If the liquid level changes, briefly open to atmosphere again.
   m. Wait until the level in the indicator tube is on zero.
   n. Slowly close the piston to empty the gas in the syringe into the absorption globe. Slowly draw the piston out again.
   o. Repeat until no significant changes in level occurs by reading the indicator tube marking – V2.

13. Repeat for CO2 flow rates of 3, 4 and 5 L/min with air flow rate set at 20 L/min.

<u>Part 2b: Desorption of CO2 into Laboratory Air</u>

1. Once CO2 values are constant, shut off the CO2 flow at the flowmeter C3, decrease the liquid flow rate to 1.5 L/min, open the drain valve D1, under the column, and close the control valve C4. Adjust drain valve D1 under the column to maintain liquid level about sample tap S4.
2. Immediately take samples as quickly as possible from S4 in -minute intervals.
3. Analyze the samples as directed in Part 2a, number 5.

<u>Part 3: Absorption of CO2 by H20 in a Continuous Column Mode</u>

1. Close drain valve under the liquid reservoir tank, the control valve C4, and the column drain D4.
2. Fill the liquid reservoir tank full of DI water.
3. Record the temperature of the water.
4. Collect a sample and analyze is as directed in Part 2a, number 5.
5. Open the control valve C4. With gas flow control valves C2 and C3 closed, start the liquid pump. Open valve C1 a couple of turns for 15-30 seconds to expel any air in the pump.
6. Adjust the water flow rate to 1.5 L/min, open valve D1 and close valve C4.
7. Use valve D1 to maintain liquid water height below the air/CO2 inlet but able the sampling valve S4.
8. Adjust the air flow rate through flowmeter F2 to 20 L/min using valve C2, adjusting as needed and recording all adjustments made.
9. Open the main valve on the CO2 cylinder all the way.
10. Adjust the pressure regulating valve on the CO2 regulator to give a gauge reading of approximately 20 psi. Adjust valve C3 to give a CO2 flow rate on flowmeter F3 to the value assigned by the TA. This will be determined during the lab period.
11. As soon as the CO2 is turned on, take a 50 mL sample from S4 at 30 second intervals during the first 2 minutes.
12. Increase the interval between samples to one minute for the next 5 minutes.
13. Increase the interval to 5 minutes until the concentration reaches a steady state.

14. Analyze the samples per Part 2a, number 5.
15. Analyze the gas samples from S2 once steady state has reached per Part 2a, number 12.

The independent variables are the time intervals the samples are taken at and the flow rates of CO_2 gas and water. The dependent variable is the amount of CO_2 in each sample.

Project Safety Evaluation (10 pts)

Commented [PTH17]: 9.5/10

Project Title: ____________Gas Absorption in a Packed Bed_________________________________

Date: _________________23 Jan 2020____

	Yes/No	
Potential electrical hazards?	Yes	Water pump is electrical. Water can get on the electrical socket posing a possible electrocution hazard.
Potential mechanical hazards?	Yes	High pressure gas cylinders
Potential pressure hazards?	Yes	High pressure gas cylinders
Potential temperature hazards?	Yes	Expected Room Temperature Joule-Thompson cooling effect of CO2 from decrease in pressure.
Hazardous/toxic chemicals involved?	Yes	NaOH
Gloves required?	Yes	Gloves are required while handling the NaOH solution
MSDS reviewed by all team members?		List MSDS sheets reviewed: Water-CAS# 7732-18-5 Carbon Dioxide P-4574-l NaOH- CAS# 1310-73-2
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	25	none
Pressure (psi)	800 psi	Appeared on the high-pressure gas cylinder. 800 psi for max on cylinder 40 psi max for process.
Safety Controls	*yes/no*	*If yes, Provide description*
	yes	On/off switches on the pump. Heater on CO2 cylinder to avoid CO2 lines clogging from Joule-Thompson effect from rapid drop in pressure. NaOH standard solutions are stored in fume hood. Shut-off valve for compressed air.

Commented [PTH18]: pump is also part of mechanical hazards

Commented [AZ19]: what is the hazard?

Commented [AZ20]: Ok

Commented [AZ21]: Good, also mention compressed air pressures.

Commented [AZ22]: Good

I have read relevant background material for the Unit Operations Laboratory entitled: "____________________Gas Absorption in a Packed Bed________________________________" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

_______Jon Chiu__________________23 Jan 2020___

signature of team member date

_______Melissa Djonlich____________23 Jan 2020___

signature of team member date

Yier Li 23 Jan 2020

signature of team member date

Cheryl Slykas 23 Jan 2020

signature of team member date

Caleb Tristano 23 Jan 2020

signature of team member date